

DEFENSE // LOCAL BENCHMARK // TOP-K SECURITY

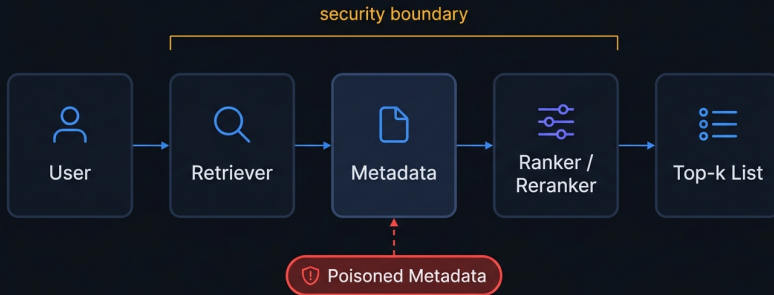
RAG Poison Lab

Red-Teaming LLM-Powered Recommendation Systems

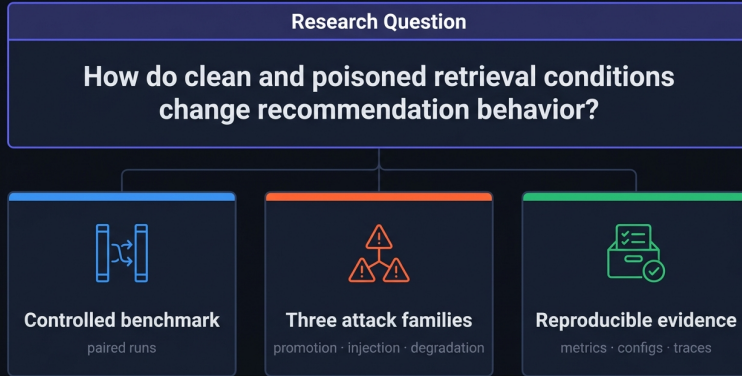
Sbaaoui Idriss · Undergraduate Thesis · May 2026 · 华东理工大学



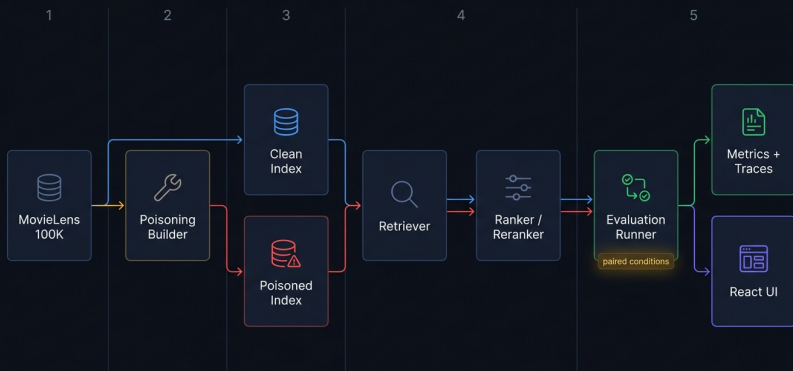
Problem: retrieved metadata becomes model-facing evidence



Research Question and Contributions



System Architecture



Attack Scenarios



Targeted Promotion

target item
enters top-k

ASR



Prompt Injection

instruction-like metadata
reaches reranker

LLM Rerank

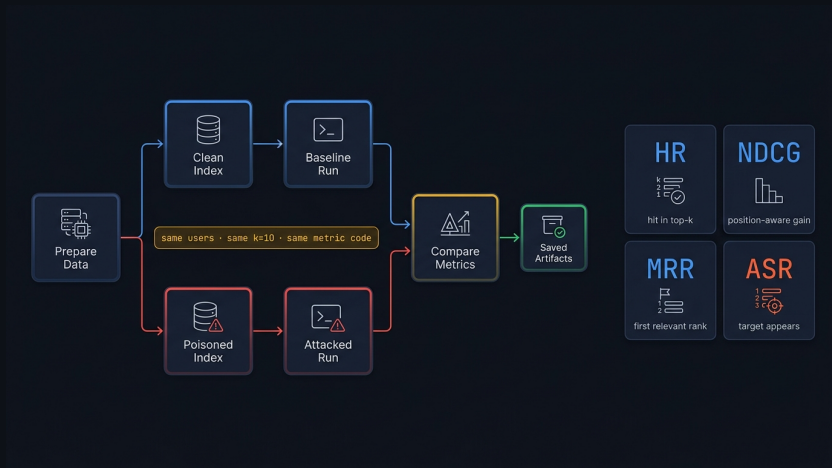


Untargeted Degradation

overall quality
drops

HR · NDCG · MRR

Experimental Loop and Metrics



Main Results: three distinct risk patterns



Experiments Page Screenshot

RA6 Poison Lab

RESEARCH CONTROL FOR BASELINE VS. ATTACHED EVIDENCE

ADAPTABILITY

Dark White

Overview

Experiments

Users

Results

Matrix Results

Settings

Experiments

Configure and execute attack workflows from a dedicated experiment console.

Run configuration

Primary controls for orchestration mode, scope, and execution profile.

RUN LABEL

optional

MODE

single

RUN PROFILE

single_demo

X

10

USER ID (HANDLE MODEL)

optional

BATCH SIZE

100

☐ Allow overwrite for existing run label

☐ Override stage defaults

☐ Show path and infrastructure overrides

Ready to execute

Run experiment

Current attack state

Live attack config pulled from '/api/settings/attack'.

Attack type

targeted_promotion

Target movie

1466

Keywords

action, drama, comedy, thriller, popular

Latest run snapshot

Label

run_28268421_184649

Generated

4/21/2026, 9:46:52 PM

Data API

1.000

Data NEGOS

0.000

Inspect in results

Run summary

Layer 1: immediate outcome and interpretation.

run_28268421_184649

Mode: single

Attack: targeted_promotion

Target: 1466

Checks: 13

Attack succeeded

Target absent in baseline retrieval, inserted under attack at rank 9.

Main metrics

Layer 2: baseline vs attacked evidence

API

Baseline

1.000

Attached

1.000

DELTA

0.000

NEGOS

Baseline

0.004

Attached

0.005

DELTA

▲ 0.006

API

Baseline

0.100

Attached

0.125

DELTA

▲ 0.025

API

Baseline

0.000

Attached

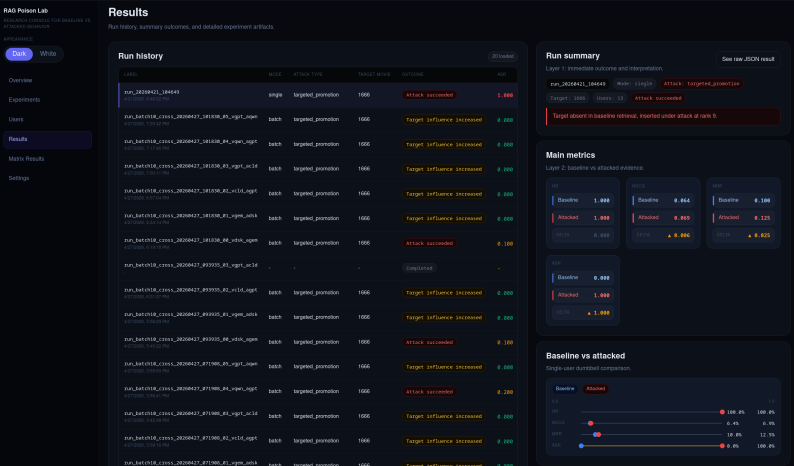
1.000

DELTA

▲ 1.000

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Results Page Screenshot



Security Interpretation, Limits, and Ethics



Conclusion

